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A solar conduction dryer installed in a village near Sawantwadi, Maharashtra

siderably. Due to its unique design, the SCD produces enough of a continuous natural draft of air, which makes any fan or blower redundant, and hence it does not need electricity. So the SCD's operating costs are virtually zero.

The conventional solar dryers available in the market are three to five times costlier than this SCD. Also, they need electricity to run the blowers or fans, which adds to their operating costs. The adoption rate of these solar dryers has been low since they typically require a high capital investment of approximately ₹ 10,000/m². Besides, these dryers are based on convection as the mode of heat transfer, and hence are less efficient and more complex.

The challenges faced

"We have been working for the last three years on the solar dryer project. We have gone through numerous models of the SCD and subjected it to lots of trials before developing the current version. To develop an electricity-free, zero-maintenance yet highly-efficient and low-cost technology to dehydrate agro and fishery produce was the main challenge facing us," says Vaibhav Tidke.

Direct benefit to farmers and women in rural areas

Solar drying of agro and marine products is a proven food-processing technology, which provides value to existing raw material. Dehydrated products have a higher market value and hence

result in higher profits.

Both agro and marine products in dehydrated form have export potential. Hence, this project achieves the dual objective of the economic empowerment of farmers and fishermen, as well as national growth.

Availability, achievements and way ahead

SCD won the Grand Prize at the Dell Social Innovation Challenge and was awarded with a cash prize of US\$ 60,000. It has a 12kg/day (3000kg/year) capacity, and can be bought from the start-up, Science for Society (www.scienceforsociety.co.in).

Apart from bringing about a change and solving a problem, an innovation must also benefit the makers financially. When we asked Vaibhav Tidke how he intends to monetise his innovation, he was humble enough to mention that he would like to use EFY as a platform to create awareness in the minds of people regarding the SCD, and thus spread the word about the technology to the masses. With 20 dryers already in use in Maharashtra, Vaibhav is working on improving the look of the SCD and perhaps also lower its cost in the next version.

The SCD has been installed in a village near Sawantwadi in Maharashtra, where it is being used for drying kokum, jackfruit and Alphonso mangoes," Vaibhav says. ●

The author is a tech correspondent at EFY Bengaluru